Comparitive Performance Report for main-ec-16K-soko07-2+1-6osd vs stretchy-C-16K-soko07-2+1-3osd vs stretchy-C-16K-soko07-2+1-6osd

Table of contents

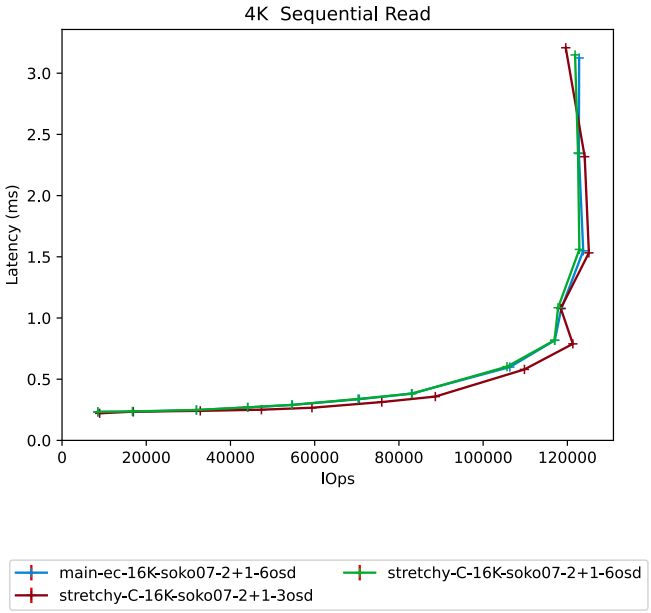
- Comparison summary for main-ec-16K-soko07-2+1-6osd vs stretchy-C-16K-soko07-2+1-3osd vs stretchy-C-16K-soko07-2+1-6osd
- Response Curves
  - Number of Jobs 1
    - Sequential Read
    - Sequential Write
    - Random Read
    - Random Write
    - Random Read/Write
- Configuration yaml files
  - results

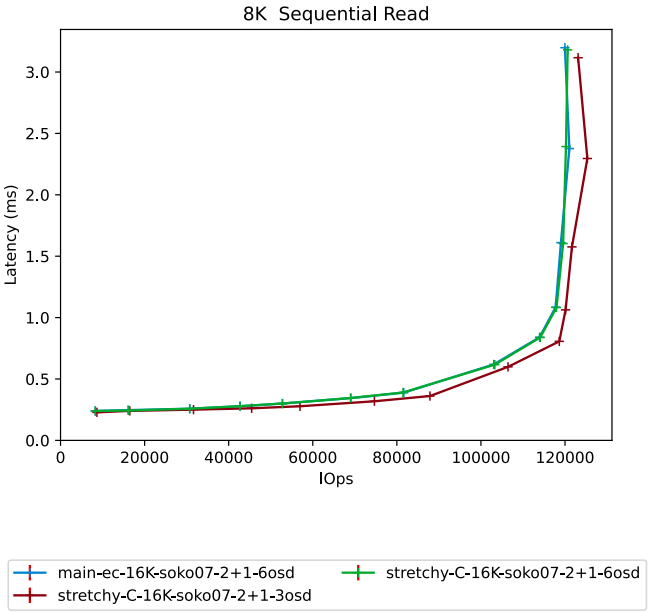
Comparison summary for main-ec-16K-soko07-2+1-6osd vs stretchy-C-16K-soko07-2+1-3osd vs stretchy-C-16K-soko07-2+1-6osd

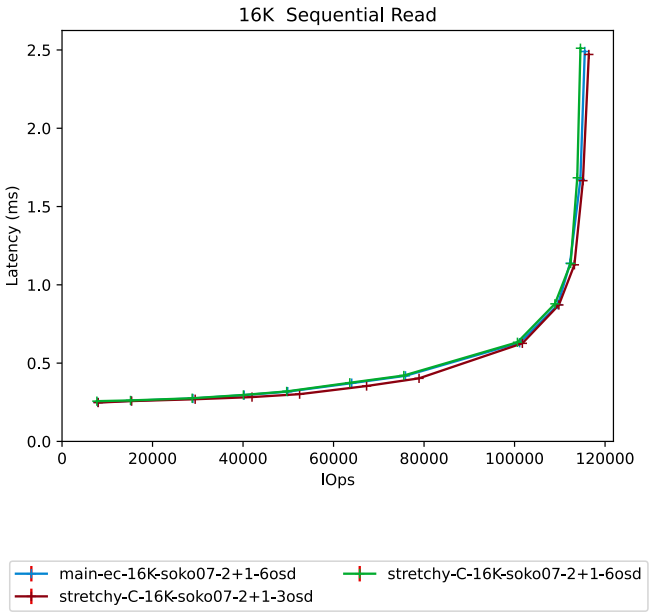
Response Curves

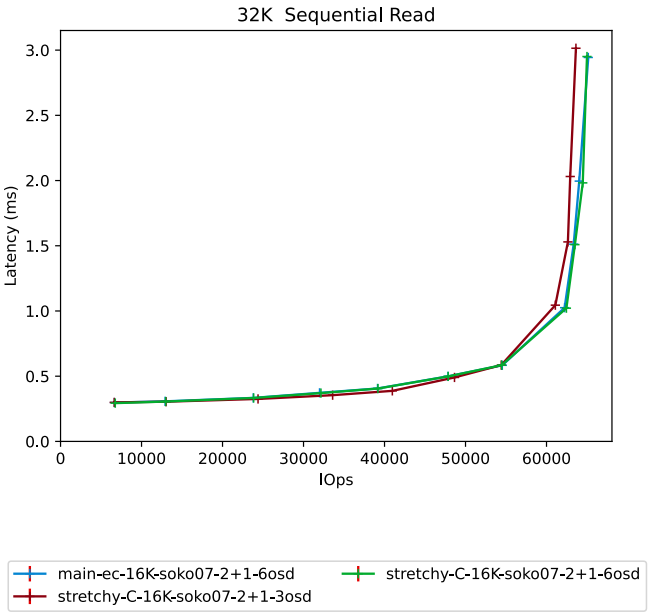
Number of Jobs 1

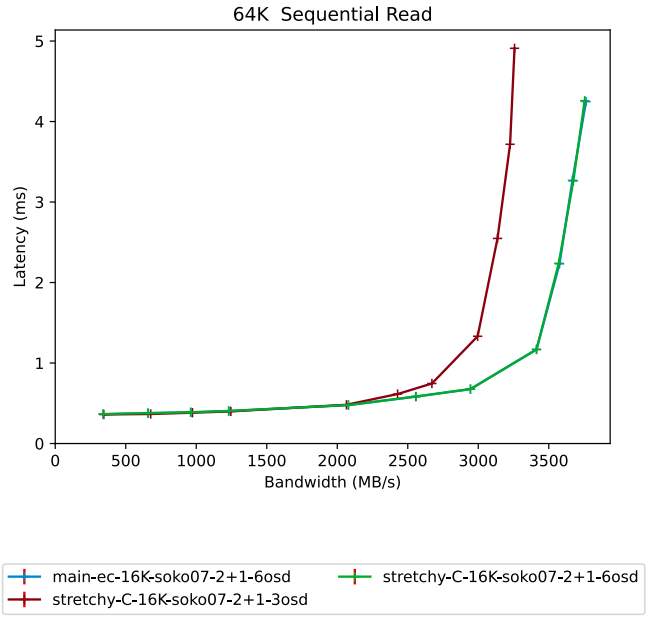
Sequential Read



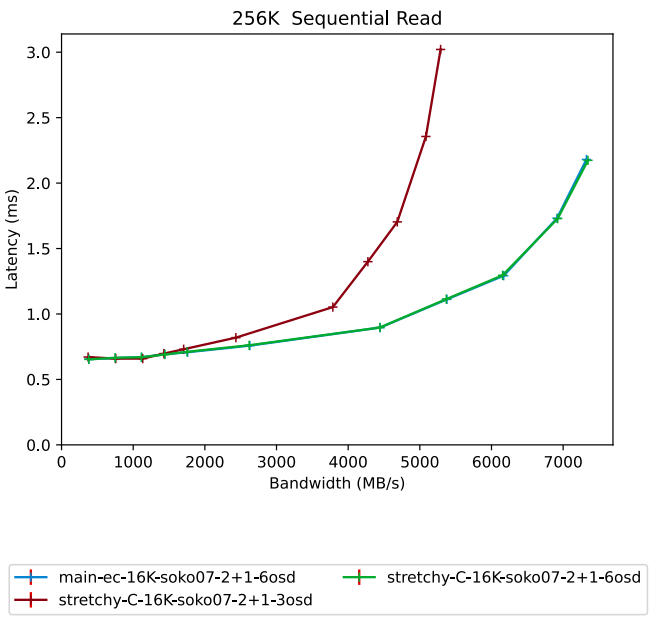




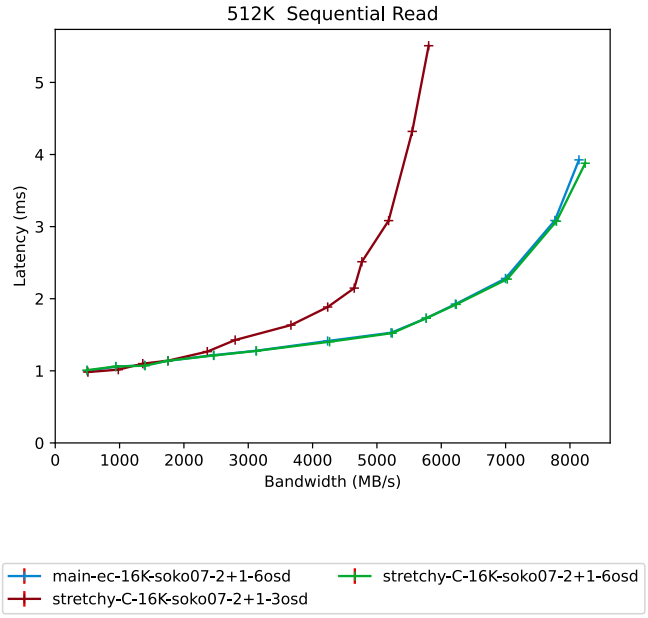



64K Sequential Read
Latency (ms)
Bandwidth (MB/s)
0 500 1000 1500 2000 2500 3000 3500
0 1 2 3 4 5

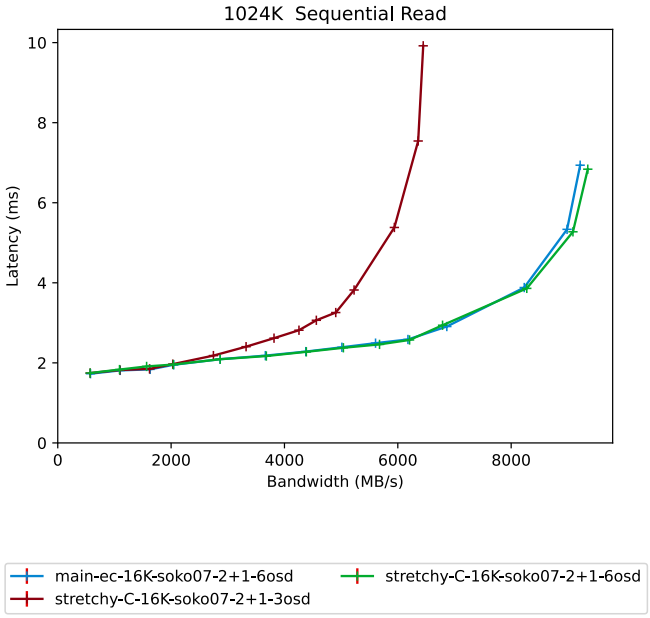
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


256K Sequential Read
Latency (ms)
Bandwidth (MB/s)
0 1000 2000 3000 4000 5000 6000 7000
0.0 0.5 1.0 1.5 2.0 2.5 3.0

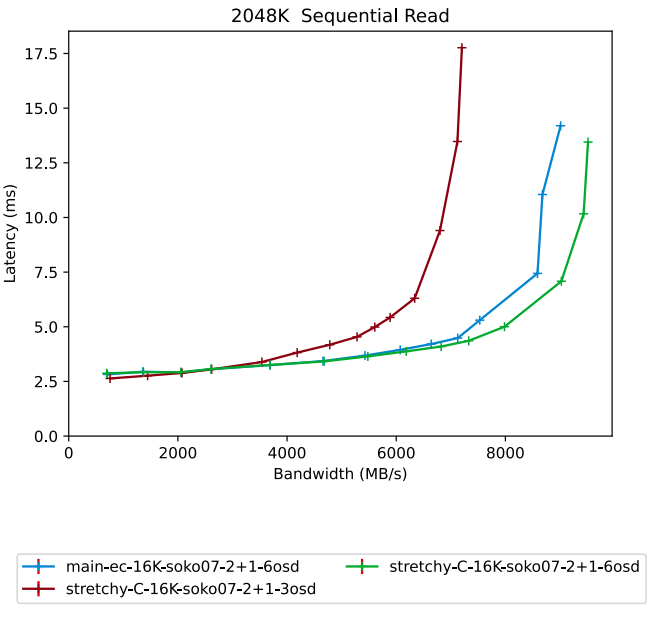
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


512K Sequential Read
Latency (ms)
Bandwidth (MB/s)
0 1000 2000 3000 4000 5000 6000 7000 8000
0 1 2 3 4 5

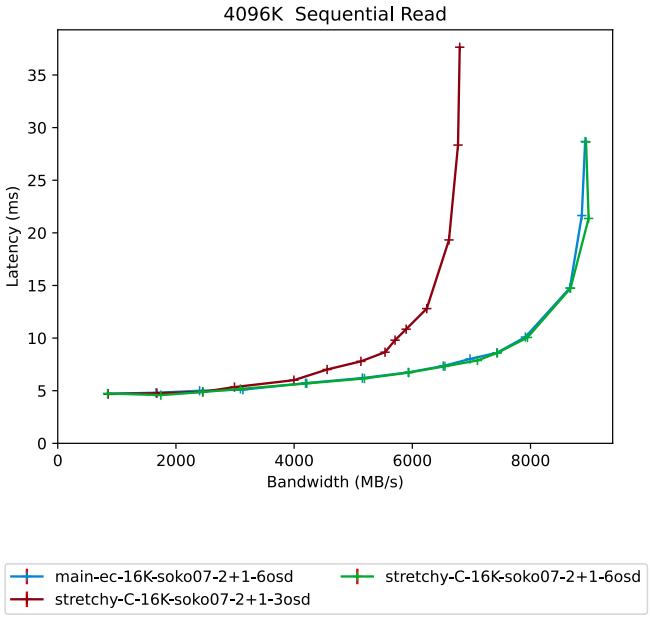
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


1024K Sequential Read
Latency (ms)
Bandwidth (MB/s)
0 2000 4000 6000 8000
0 2 4 6 8 10

main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

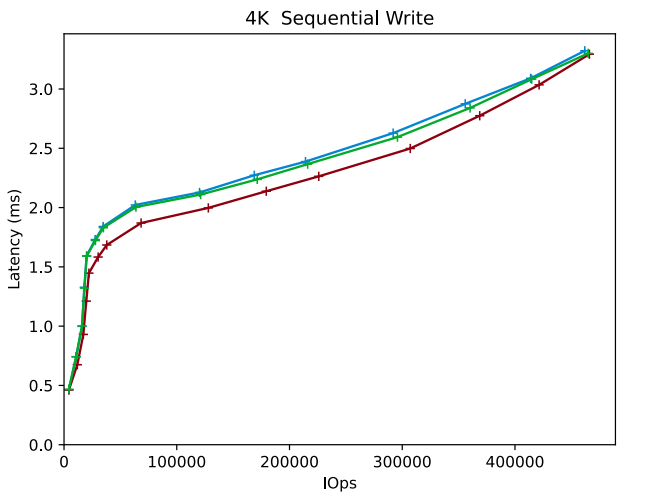

2048K Sequential Read
Latency (ms)
Bandwidth (MB/s)
0 2000 4000 6000 8000
0.0 2.5 5.0 7.5 10.0 12.5 15.0 17.5

main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

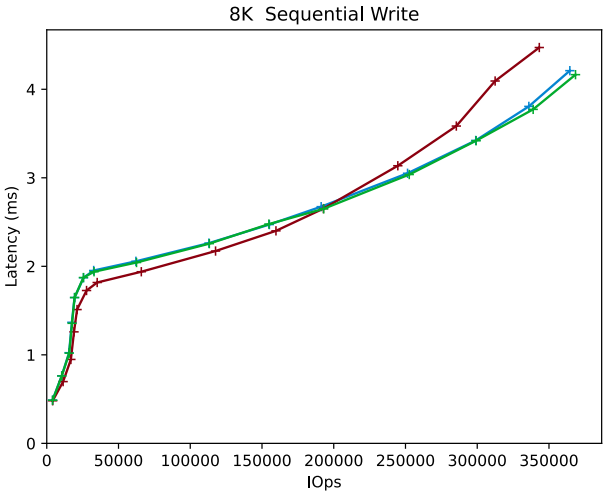

4096K Sequential Read
Latency (ms)
Bandwidth (MB/s)
0 2000 4000 6000 8000
0 5 10 15 20 25 30 35

main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

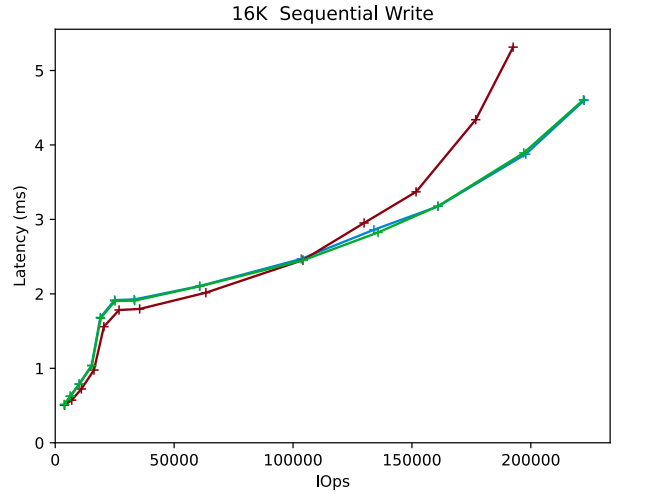
Sequential Write



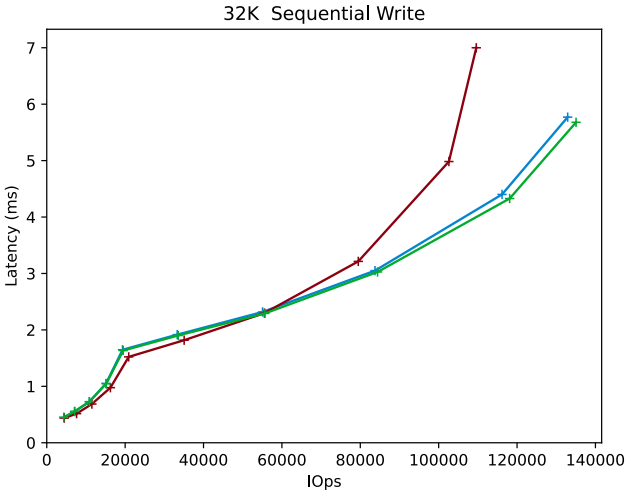




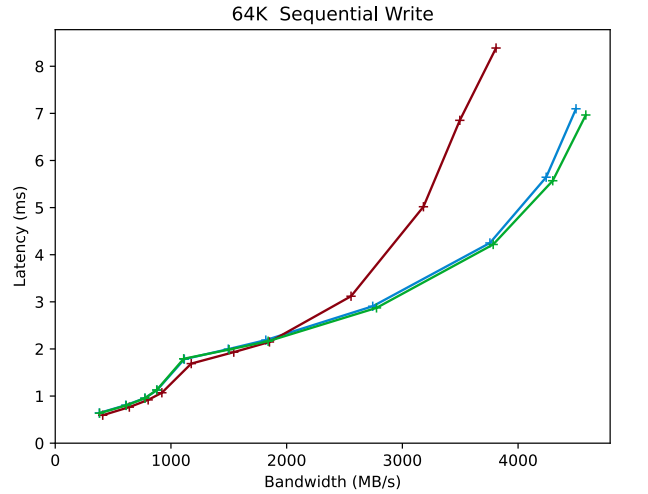




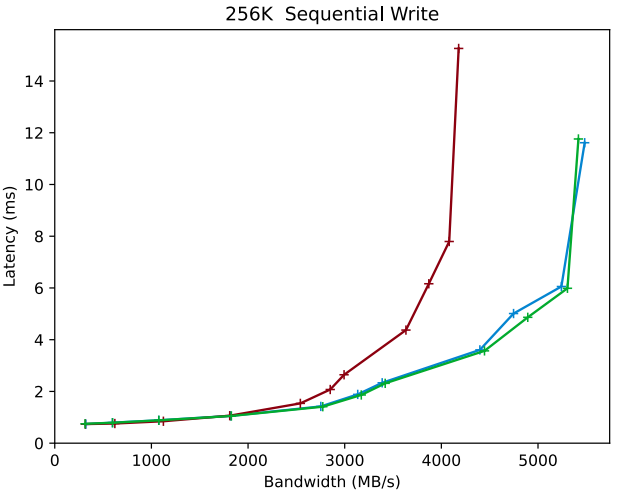




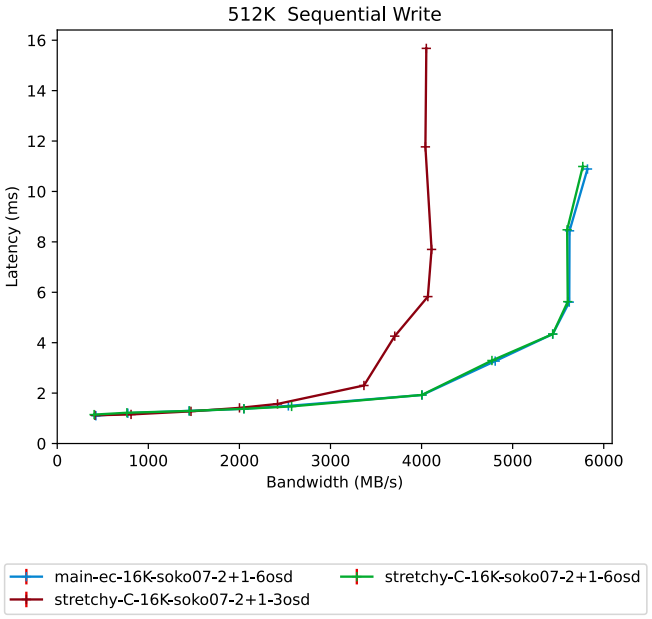


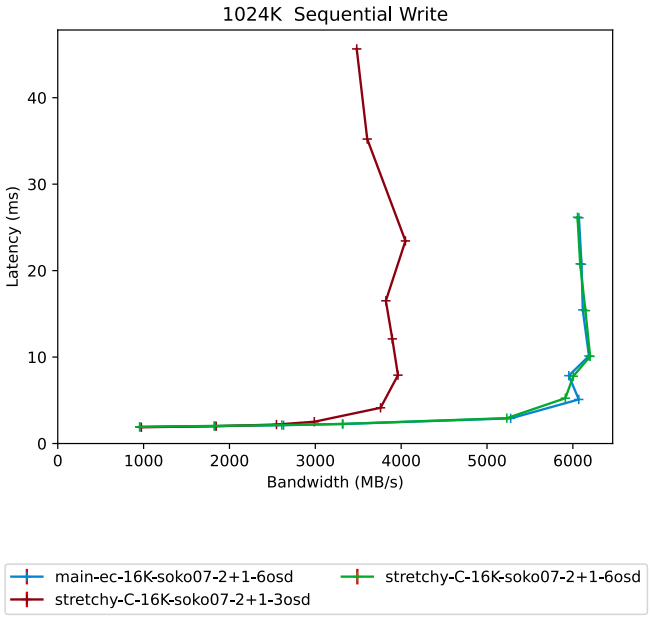


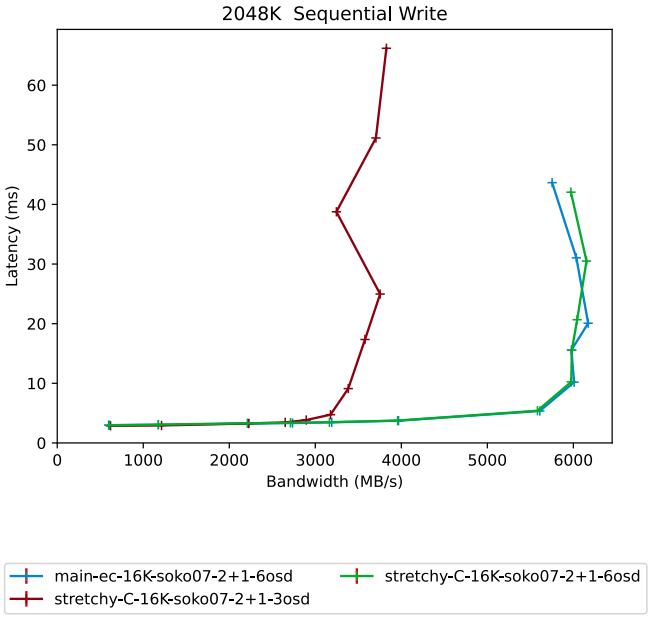


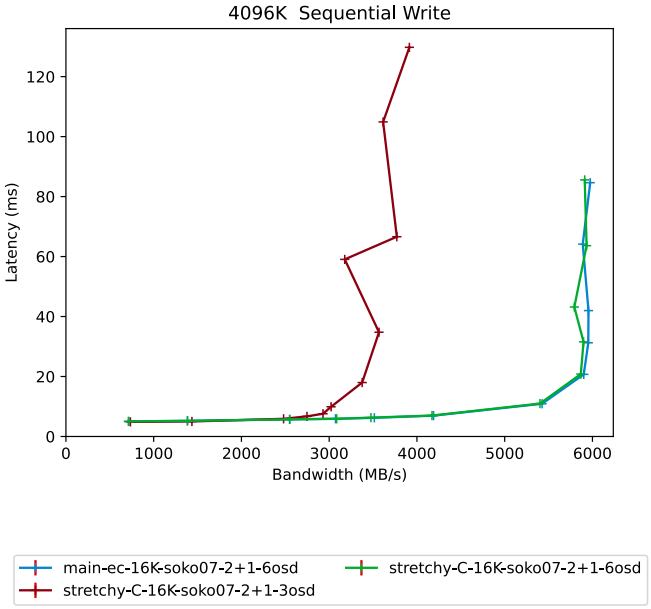




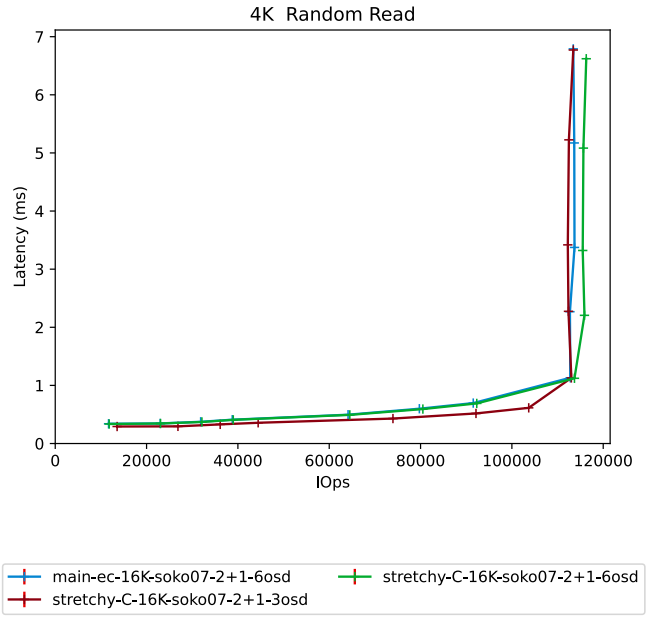

512K Sequential Write
Latency (ms)
Bandwidth (MB/s)
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


1024K Sequential Write
Latency (ms)
Bandwidth (MB/s)
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

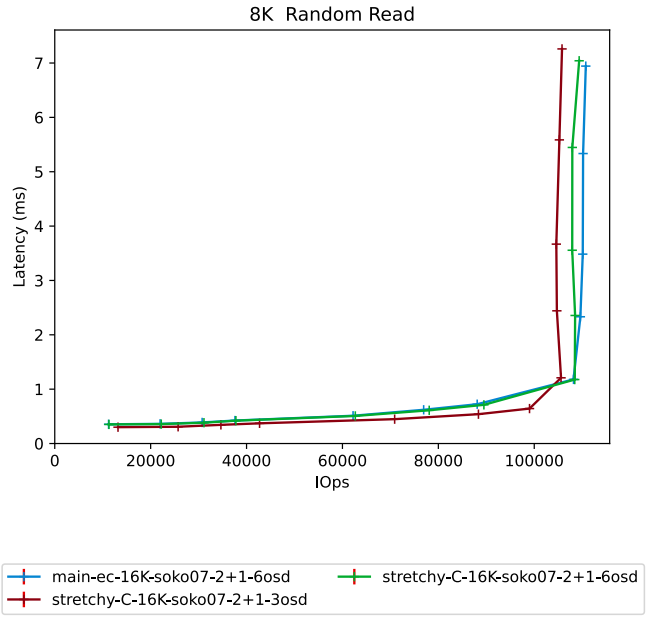

2048K Sequential Write
Latency (ms)
Bandwidth (MB/s)
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


4096K Sequential Write
Latency (ms)
Bandwidth (MB/s)
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

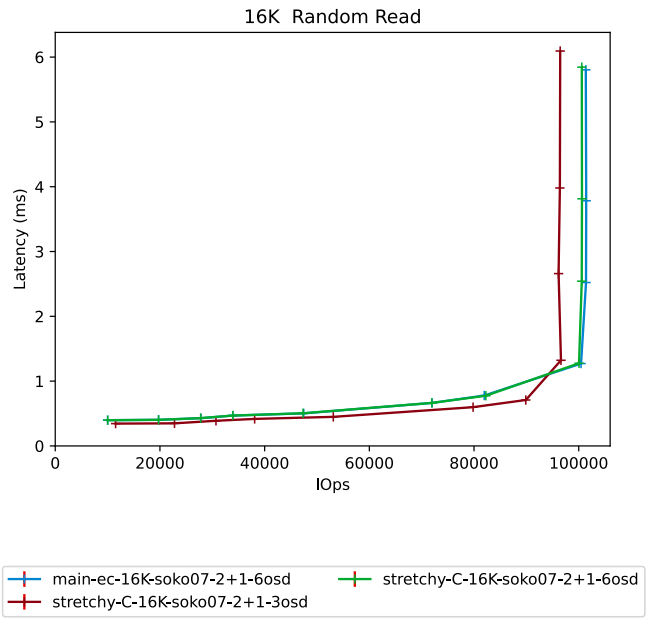
Random Read


4K Random Read
Latency (ms)
IOps
0
20000
40000
60000
80000
100000
120000
1
2
3
4
5
6
7

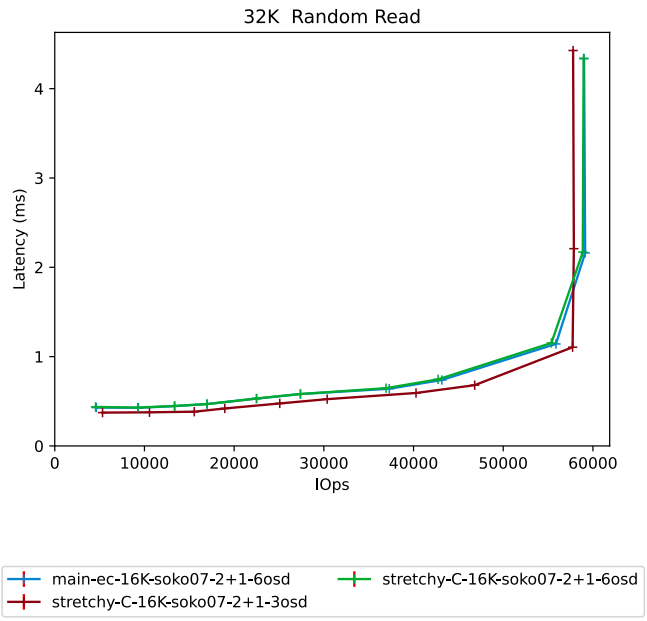
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


8K Random Read
Latency (ms)
IOps
0
20000
40000
60000
80000
100000
1
2
3
4
5
6
7

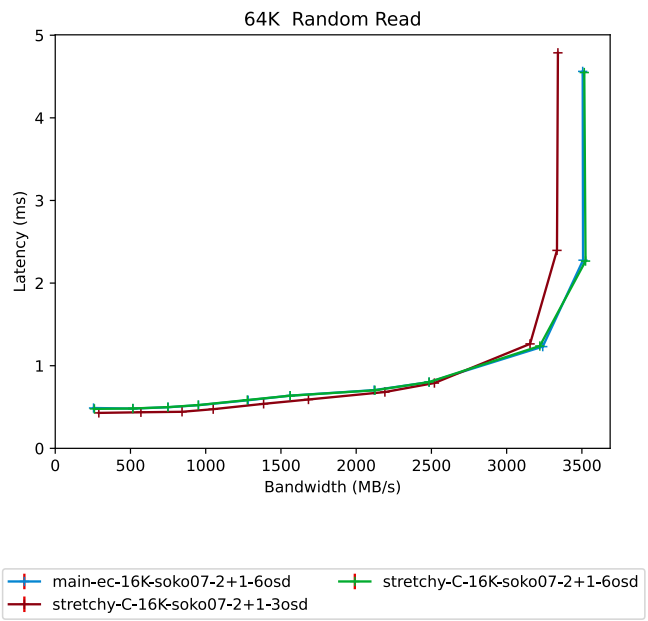
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


16K Random Read
Latency (ms)
IOps
0
20000
40000
60000
80000
100000
1
2
3
4
5
6

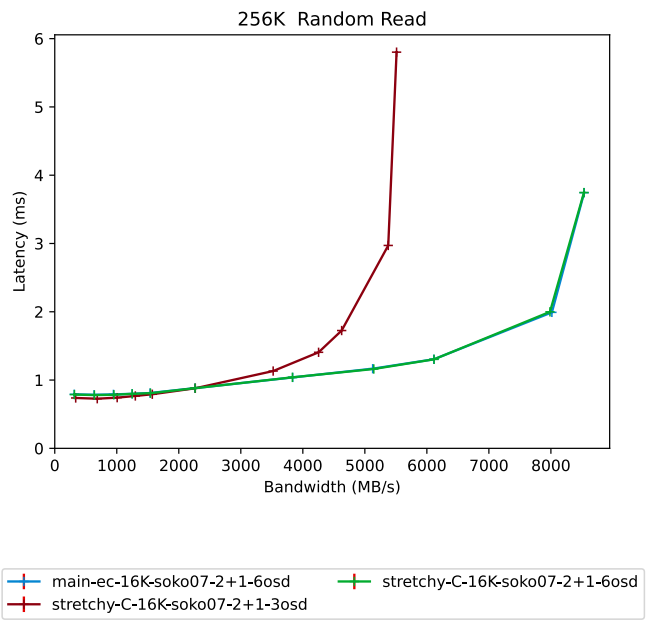
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


32K Random Read
Latency (ms)
IOps
0
10000
20000
30000
40000
50000
60000
1
2
3
4

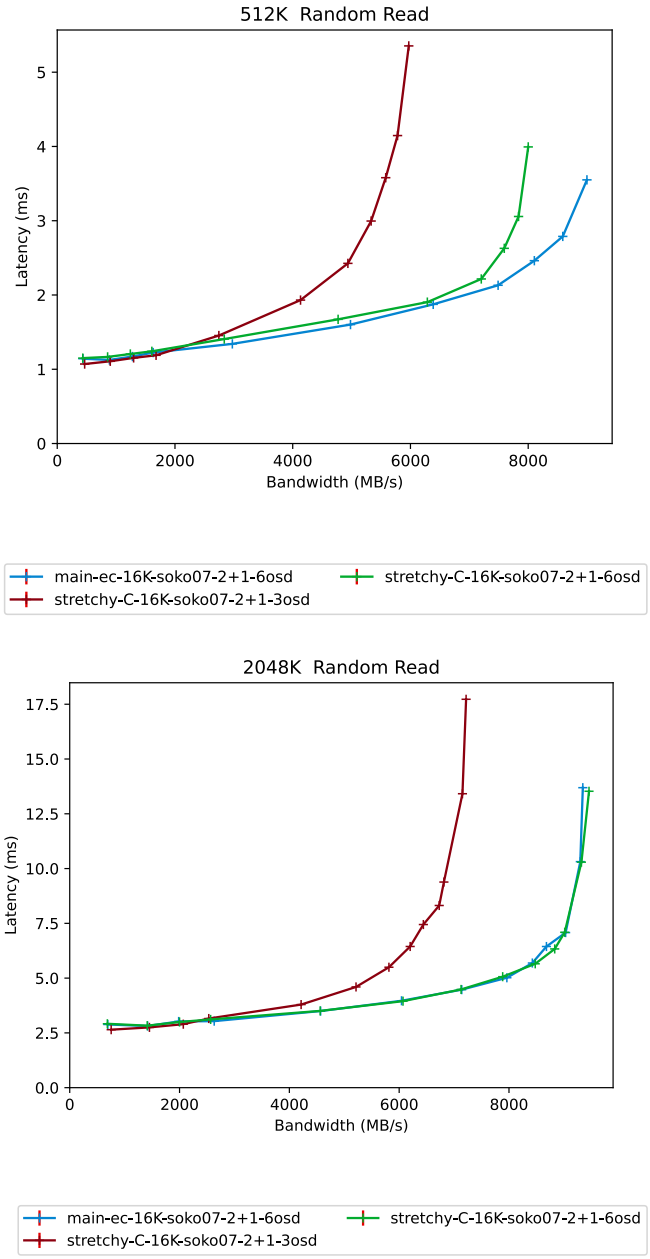
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

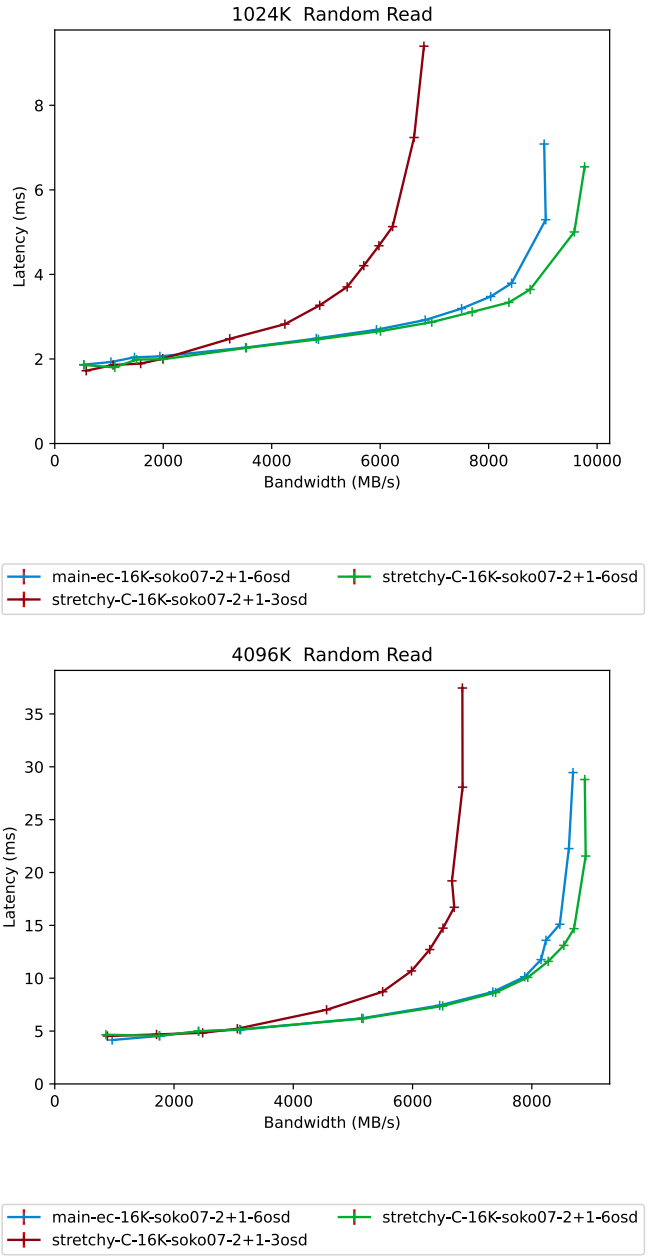

64K Random Read
Latency (ms)
Bandwidth (MB/s)
0
500
1000
1500
2000
2500
3000
3500
1
2
3
4
5

main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

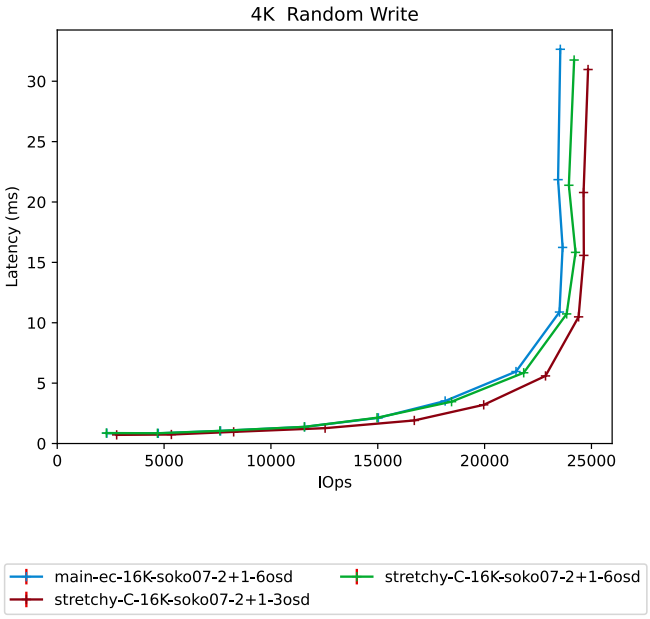

256K Random Read
Latency (ms)
Bandwidth (MB/s)
0
1000
2000
3000
4000
5000
6000
7000
8000
1
2
3
4
5
6

main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

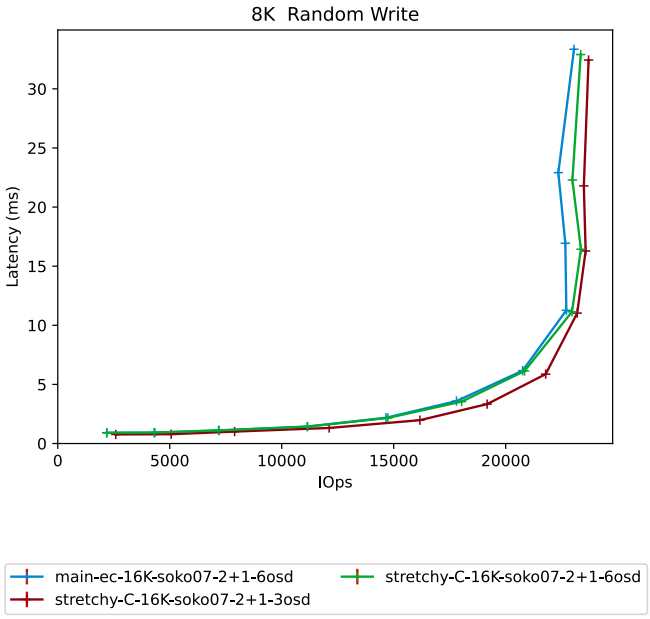




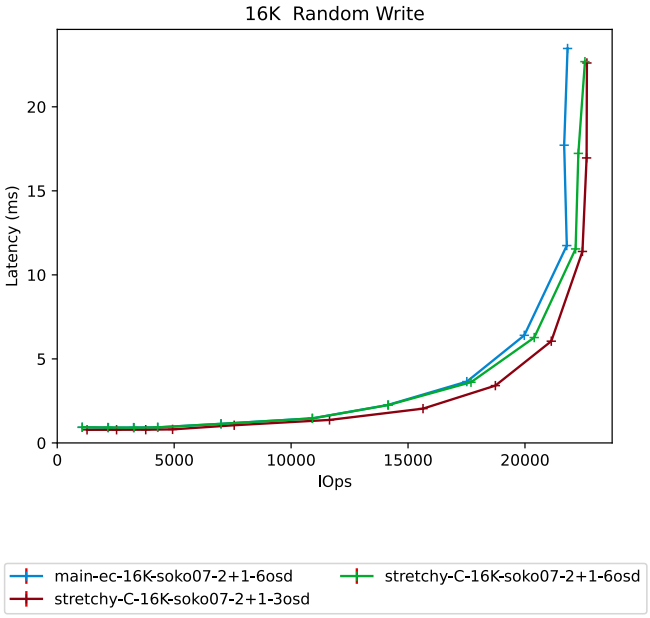
Random Write


4K Random Write
Latency (ms)
IOps
0
5
10
15
20
25
30
0
5000
10000
15000
20000
25000

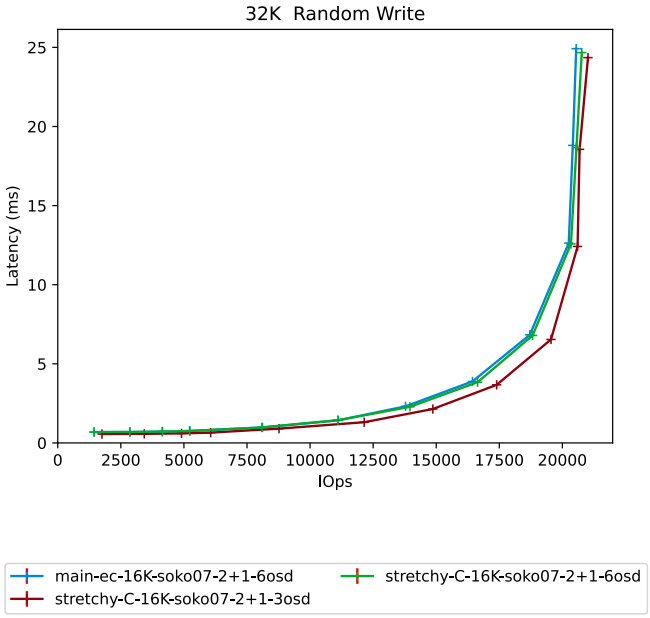
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


8K Random Write
Latency (ms)
IOps
0
5
10
15
20
25
30
0
5000
10000
15000
20000

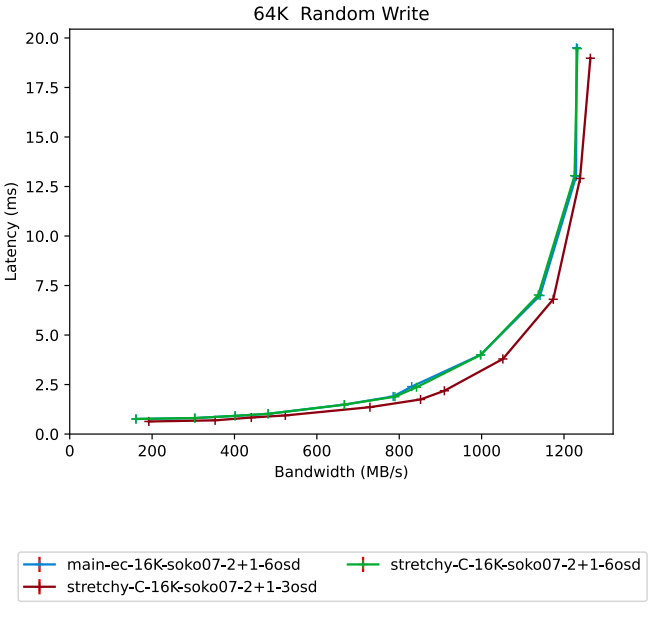
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


16K Random Write
Latency (ms)
IOps
0
5
10
15
20
0
5000
10000
15000
20000

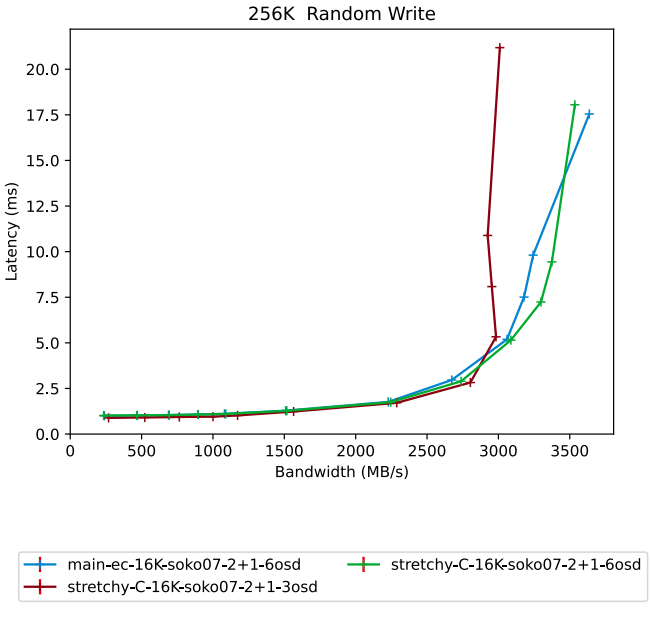
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


32K Random Write
Latency (ms)
IOps
0
5
10
15
20
25
0
2500
5000
7500
10000
12500
15000
17500
20000

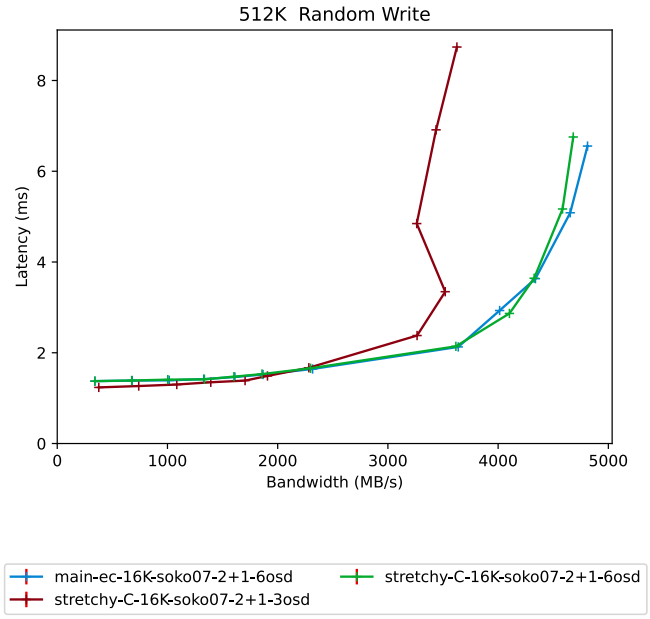
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

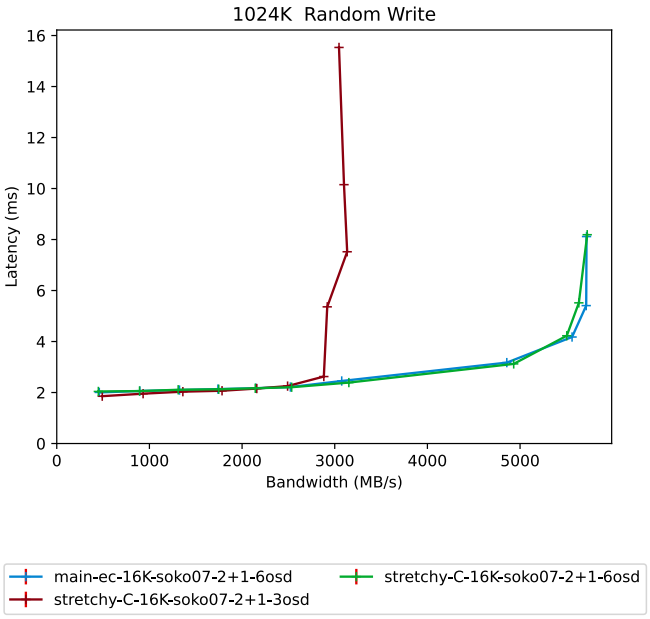

64K Random Write
Latency (ms)
Bandwidth (MB/s)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
200
400
600
800
1000
1200

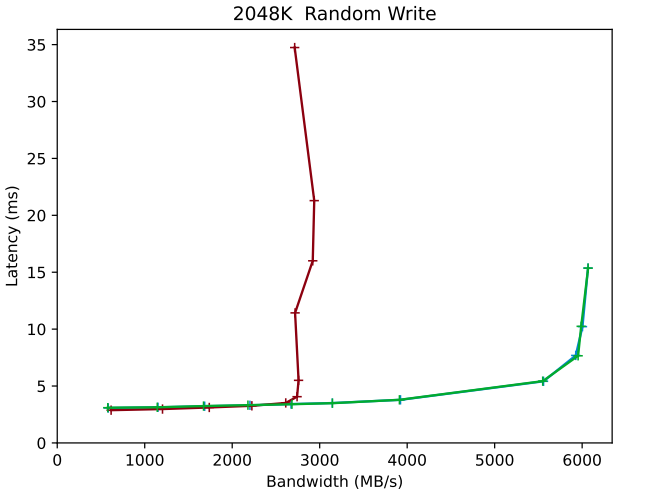
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

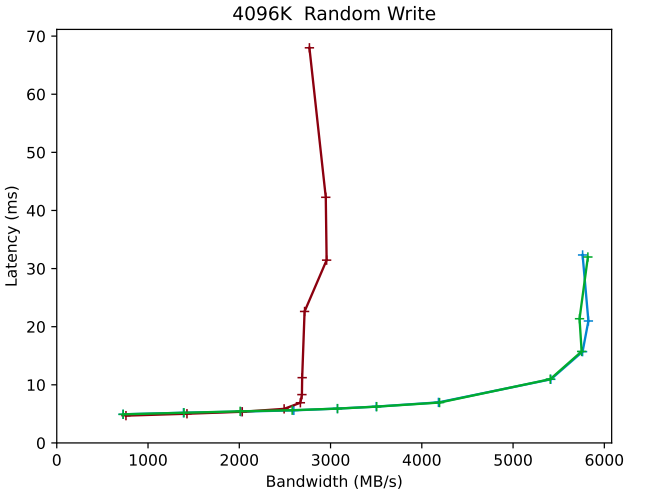

256K Random Write
Latency (ms)
Bandwidth (MB/s)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
500
1000
1500
2000
2500
3000
3500

main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd





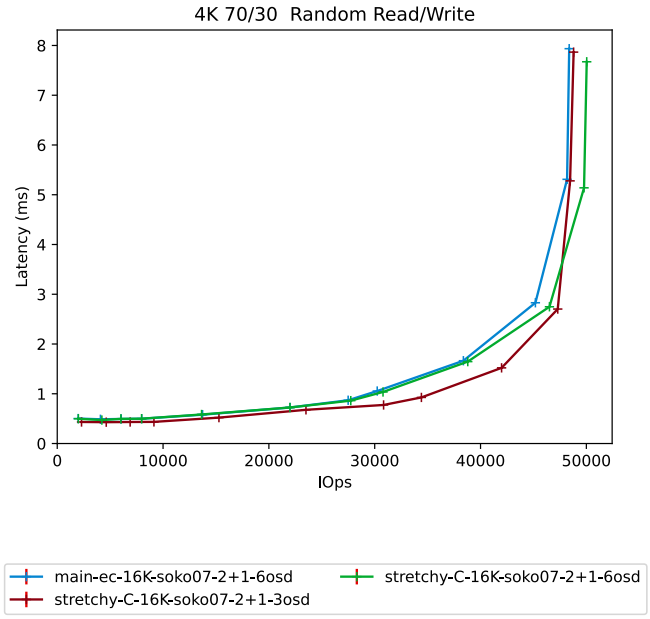




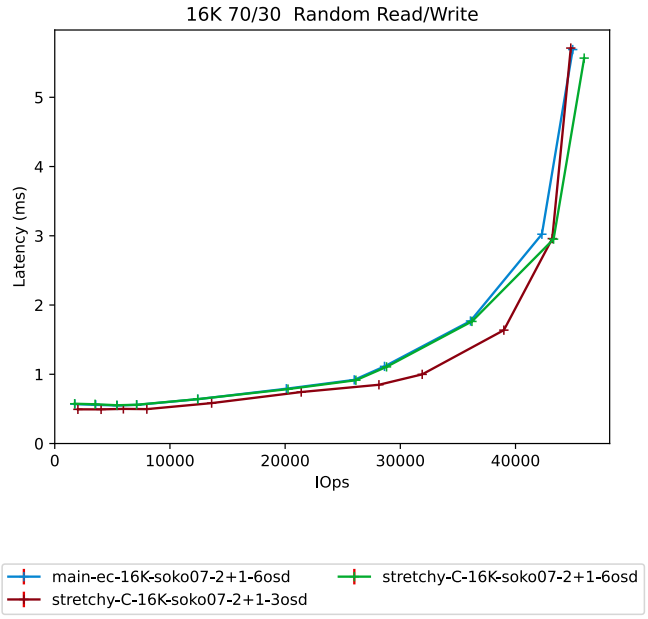




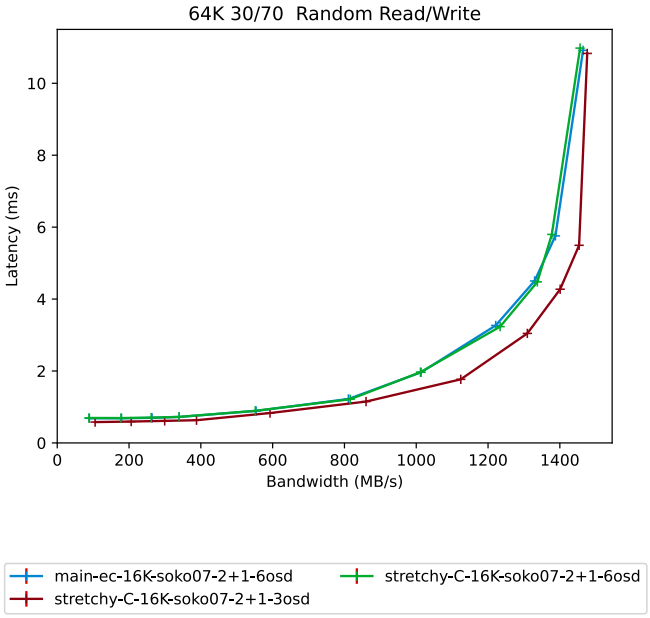
Random Read/Write


4K 70/30 Random Read/Write
Latency (ms)
IOps
0
1
2
3
4
5
6
7
8
0
10000
20000
30000
40000
50000

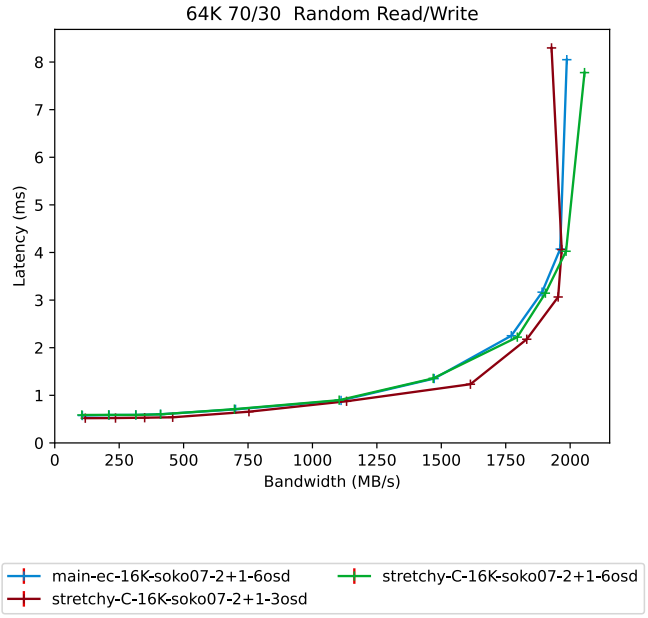
main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


16K 70/30 Random Read/Write
Latency (ms)
IOps
0
1
2
3
4
5
0
10000
20000
30000
40000

main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


64K 30/70 Random Read/Write
Latency (ms)
Bandwidth (MB/s)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd


64K 70/30 Random Read/Write
Latency (ms)
Bandwidth (MB/s)
0
1
2
3
4
5
6
7
8
0
250
500
750
1000
1250
1500
1750
2000

main-ec-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-6osd
stretchy-C-16K-soko07-2+1-3osd

Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results


```
librbdfio:
  cmd_path: /usr/local/bin/fio
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
  - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
  - 1
  ramp: 20
  rbdname: cbt-librbdfio
  time: 60
  time_based: true
  use_existing_volumes: true
  vol_size: 185625
  volumes_per_client:
  - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    16k7030:
      jobname: randmix
      mode: randrw
      numjobs:
      - 1
      op_size: 16384
      rwmixread: 70
      total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
    16krandomread:
      jobname: randread
      mode: randread
      numjobs:
      - 1
      op_size: 16384
      total_iodepth:
      - 4
      - 8
      - 12
      - 16
      - 24
      - 48
      - 64
      - 128
      - 256
```

```
  - 384
  - 588
16krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 16384
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 128
  - 256
  - 384
  - 512
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
  - 48
  - 64
1Mrandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 6
  - 8
  - 16
  - 24
  - 32
  - 48
1Mseqread:
  jobname: seqread
  mode: read
  numjobs:
```

```
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 6
  - 8
  - 10
  - 12
  - 14
  - 16
  - 20
  - 32
  - 48
  - 64
1Mseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 16
  - 32
  - 48
  - 64
  - 96
  - 128
  - 160
256krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
256krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 262144
  total_iodepth:
  - 1
  - 2
  - 3
```

```
    - 4
    - 5
    - 8
    - 16
    - 32
    - 64
    - 96
    - 128
    - 256
  2Mrandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 2097152
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 48
    - 64
  2Mrandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 2097152
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 6
    - 8
    - 16
    - 24
    - 32
    - 48
  2Mseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 2097152
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 10
    - 12
    - 14
```

```
  - 16
  - 20
  - 32
  - 48
  - 64
2Mseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 2097152
  total_iodepth:
  - 1
  - 2
  - 4
  - 5
  - 6
  - 8
  - 16
  - 32
  - 48
  - 64
  - 96
  - 128
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 32768
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
32krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 32768
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 128
  - 256
  - 384
  - 512
4Mrandomread:
  jobname: randread
```

```
    mode: randread
    numjobs:
    - 1
    op_size: 4194304
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 48
    - 64
  4Mrandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 4194304
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 6
    - 8
    - 16
    - 24
    - 32
    - 48
  4Mseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 4194304
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 10
    - 12
    - 14
    - 16
    - 20
    - 32
    - 48
    - 64
  4Mseqwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 4194304
```

```
  total_iodepth:
  - 1
  - 2
  - 4
  - 5
  - 6
  - 8
  - 16
  - 32
  - 48
  - 64
  - 96
  - 128
4k7030:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 4096
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 384
4krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 4096
  total_iodepth:
  - 4
  - 8
  - 12
  - 16
  - 32
  - 48
  - 64
  - 128
  - 256
  - 384
  - 588
  - 768
4krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 4096
  total_iodepth:
  - 2
  - 4
  - 8
  - 16
```

```
    - 32
    - 64
    - 128
    - 256
    - 384
    - 512
    - 768
  512krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 524288
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
  512krandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 524288
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 6
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
  512kseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 524288
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 12
    - 16
    - 20
    - 24
    - 32
    - 48
```

```
  - 64
512kseqwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 524288
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 16
  - 32
  - 48
  - 64
  - 96
  - 128
64k3070:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 30
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 96
  - 128
  - 256
64k7030:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 96
  - 128
  - 256
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
```

```
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
64krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 384
64kseqread:
  jobname: read
  mode: read
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 192
  - 256
64kseqwrite:
  jobname: write
  mode: write
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 4
  - 8
  - 12
  - 16
  - 32
  - 48
  - 64
  - 128
```

```
  - 256
  - 384
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 8192
  total_iodepth:
  - 4
  - 8
  - 12
  - 16
  - 32
  - 48
  - 64
  - 128
  - 256
  - 384
  - 588
  - 768
8krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 8192
  total_iodepth:
  - 2
  - 4
  - 8
  - 16
  - 32
  - 64
  - 128
  - 256
  - 384
  - 512
  - 768
precondition:
  jobname: precond1rw
  mode: randwrite
  monitor: false
  numjobs:
  - 1
  op_size: 65536
  time: 120
  total_iodepth:
  - 16
seq16kread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 16384
  total_iodepth:
  - 2
  - 4
  - 8
  - 12
  - 16
  - 24
```

```
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
  seq16kwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 16384
    total_iodepth:
    - 2
    - 4
    - 8
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 512
    - 768
    - 1024
  seq256kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
  seq256kwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 4
    - 8
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 256
  seq32kread:
```

```
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 32768
  total_iodepth:
  - 2
  - 4
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 96
  - 128
  - 192
seq32kwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 32768
  total_iodepth:
  - 2
  - 4
  - 8
  - 16
  - 32
  - 64
  - 128
  - 256
  - 512
  - 768
seq4kread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 4096
  total_iodepth:
  - 2
  - 4
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 96
  - 128
  - 192
  - 288
  - 384
seq4kwrite:
  jobname: seqwrite
  mode: write
  numjobs:
  - 1
  op_size: 4096
  total_iodepth:
  - 2
  - 8
```

```
        - 16
        - 24
        - 32
        - 48
        - 64
        - 128
        - 256
        - 384
        - 512
        - 768
        - 1024
        - 1280
        - 1536
      seq8kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 8192
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
      seq8kwrite:
        jobname: seqwrite
        mode: write
        numjobs:
        - 1
        op_size: 8192
        total_iodepth:
        - 2
        - 8
        - 16
        - 24
        - 32
        - 48
        - 64
        - 128
        - 256
        - 384
        - 512
        - 768
        - 1024
        - 1280
        - 1536
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
```

```
  - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:
      a: null
  mkfs_opts: -f -i size=2048
  mons:
    --- server1 ---:
      a: --- IP Address ---:6789
  mount_opts: -o inode64,noatime,logbsize=256k
  osds:
  - --- server1 ---
  osds_per_node: 8
  pdsh_cmd: /bin/pdsh
  pdsh_ssh_args: -a -x -l%u %h
  pid_dir: /tmp/cbt/pid
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -oz -sZC -i 10 --sep ";" -F0 -P -f {collectl_dir}
```